

Selection over Scale in Transformer Length Generalization

Anonymous submission

Abstract

A transformer that has learned a length-invariant rule on short sequences often fails to apply it on longer ones. We ask which internal property of the model supports generalization to longer sequences, and we answer by direct measurement. On controlled retrieval tasks the input position that holds the answer is known, so we can measure how much attention the model places on the correct source token at test length. This quantity predicts per-token accuracy with correlation 0.97, across tasks and across positional encodings. When attention stays concentrated on the correct source as competitors grow, the model generalizes, and when it disperses, accuracy falls with it. We compare this against a natural alternative, the scale of the attention output, which also degrades with length. Attention-output variance predicts accuracy less well, with correlation 0.59, and an intervention that provably holds it constant across length does not improve generalization, and lowers accuracy under rotary position embeddings. A short argument explains the gap. The attention output is a convex combination of value vectors, so its variance shrinks whenever attention spreads over more competitors, for a correct or an incorrect selection alike, while only the selection controls the behavior. We confirm the direction by intervention. Raising attention on the correct source raises accuracy in proportion, with the accuracy gain and attention gain tracking each other across the four cells (correlation 1.0), while stabilizing the variance raises neither. Attention concentration on the correct source is the operative variable that supports length generalization. The findings reproduce at eight layers and width 512, and the same pattern appears in a pretrained language model, Pythia-1.4B, on in-context retrieval, where a logit-lens decomposition splits the failure into an attention locus and a downstream readout locus. We release the code, the result files, and the pre-registration.

1 Introduction

A transformer can learn a rule on short sequences and then apply it wrongly on longer ones, even when the rule itself does not depend on length. Retrieving a marked value, selecting the entry with the largest key, and looking up a value by its key are all length-invariant, yet a model that solves them at the training length degrades steadily as the input grows. We

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

study what internal property of a trained model determines whether it keeps solving such a task at longer lengths.

For a retrieval task the requirement is mechanical. The model produces the answer by reading a value out of the sequence, so at the query position it must place attention on the token that holds that value. If attention on the correct source token stays concentrated as the number of competing tokens grows, the readout receives its input and the answer is correct. If attention spreads across the sequence, the readout receives a blend of many values and the answer degrades. This suggests a direct hypothesis. The operative variable for length generalization is how much attention the model places on the correct source token at test length.

We can test this hypothesis without inference from proxies. In each of our tasks the input position that holds the answer is fixed by construction, so we read the attention weight on that exact position at the query token, for every length, and relate it to accuracy. This gives a direct measurement of the proposed operative variable instead of a surrogate.

Several internal statistics degrade with length, and any of them is a candidate explanation. One natural candidate is the scale of the activations. The variance of the attention output shrinks as sequences lengthen, which shifts the input distribution of every downstream component, and a layer normalization on the attention output restores that scale (Li, Boduljak, and Zhou 2025). Attention entropy and attention sharpness are further candidates that change with length (Wang et al. 2024; Nakanishi 2025; Vasylenko et al. 2025). We take the scale candidate as the sharpest comparison, because it comes with a concrete intervention that provably repairs the statistic, so we can ask whether repairing it changes the behavior.

Our study crosses two axes that a single-statistic view can leave fixed. The first is the positional encoding. We test both the encoding-free setting (NoPE) (Kazemnejad et al. 2023) and rotary position embeddings (RoPE) (Su et al. 2021), because the choice interacts with components placed after attention (Hua et al. 2024; Zsámboki et al. 2025). The second is whether the task is order-dependent. We include reversed decimal addition, where token position carries information, as a contrast to the order-invariant retrieval tasks.

Our findings are as follows.

- Attention on the correct source token predicts per-token accuracy with a within-cell correlation of 0.97, across both tasks and both positional encodings. It starts near

1.0 at short length and falls toward chance as competitors grow, and accuracy tracks it.

- The attention-output variance also collapses with length, and it predicts accuracy less well, with correlation 0.59. An intervention that holds the downstream variance constant across length does not improve length generalization, and it lowers accuracy under RoPE.
- Raising attention on the correct source by inference-time logit sharpening raises accuracy in proportion. Across the four task and encoding cells the accuracy gain and the attention gain track each other at correlation 1.0, while the variance-stabilizing intervention raises neither.
- Patching the attention distribution directly onto the correct source, holding the weights fixed, restores accuracy from 0.59 to near 1.0 at $50\times$ the training length, which shows attention on the correct source is sufficient to drive the behavior.
- A convex-combination argument explains why. Variance collapse follows from dispersion for any selection, so it co-varies with failure while attention concentration controls it.
- The variance-stabilizing intervention lowers attention on the correct source under RoPE, which accounts for the accuracy it costs there.

That attention concentration matters for length generalization has been observed before (Wang et al. 2024; Vasylenko et al. 2025). Our contribution is to measure it directly against the token a retrieval task requires, to compare it head to head with the attention-output-variance candidate under a controlled task and positional-encoding grid, and to show that an intervention which repairs the variance does not transfer to the behavior and reverses under RoPE. This turns a plausible correlate into a tested one and separates it from a statistic that co-varies with the same dispersion.

2 Candidate internal correlates of length generalization

Length generalization has been approached through positional encoding (Kazemnejad et al. 2023), attention structure, and intermediate supervision (Anil et al. 2022), and it is attainable while remaining fragile to seed and data order (Zhou et al. 2024). We organize the internal-mechanism work as a set of candidate correlates, each a quantity that changes with length and could in principle drive the failure.

Attention concentration. As the number of competing key positions grows, a softmax with bounded logits spreads its mass more thinly, and the weight on any single position decays. Wang et al. (2024) relate the length at which attention entropy inflects to the length at which generalization degrades, and report a sharpening intervention that helps in the encoding-free setting. Scalable-softmax multiplies attention logits by a factor that grows with length and improves long-context retrieval (Nakanishi 2025). Sparse attention that caps the number of attended positions preserves concentration and reaches far longer sequences (Vasylenko et al. 2025). Attention entropy has also been studied as a quantity tied to

training stability (Zhai et al. 2023). These works point at the sharpness of the attention distribution. We make the target precise for retrieval by measuring attention on the known correct source, which is the specific quantity the readout depends on.

Activation scale. A second candidate is the scale of activations downstream of attention. The variance of the attention output shrinks with length, which shifts the input distribution of the components that follow, and a post-attention layer normalization restores the scale (Li, Boduljak, and Zhou 2025). This candidate is attractive to test because it names a concrete intervention that provably repairs the statistic, which lets us separate repairing the statistic from changing the behavior. The limits of normalization inside attention have themselves been examined, with softmax selection drifting toward uniform as more tokens compete (Mudarisov et al. 2025).

Positional encoding and downstream components. We keep the positional encoding as a variable. Hua et al. (2024) show that linear layers and pointwise activations act as spectral filters that damage the periodic structure RoPE relies on, which harms length generalization. Zsámboki et al. (2025) study how a post-attention normalization reshapes attention sharpness. Both bear on any intervention placed after attention, and both are relevant to the positional-encoding interaction we report.

3 Setup

3.1 Tasks

We study three synthetic tasks with exact supervision, chosen so that the answer’s dependence on token order is controlled.

Argmax retrieval presents key-value pairs and asks for the value paired with the largest key. The task is order-invariant.

Flag retrieval presents a sequence of values, exactly one of which carries a mark, and asks for the marked value. The task is order-invariant and single-hop. A small model masters it at the training length with exact-match 1.00 and degrades smoothly with length, which gives the dynamic range to read an effect.

Reversed decimal addition presents two integers and asks for their sum in least-significant-digit-first order. The task is order-dependent, because a digit’s position selects its place value, and it serves as the contrast to the retrieval tasks. Length generalization on addition is known to depend on how digit position is represented (McLeish et al. 2024).

Symbols are drawn from vocabularies large enough that long sequences contain distinct tokens, and answers are supervised token by token. Dictionary lookup, a further order-invariant retrieval task, does not reach our training-competence gate at this model scale, and we report it as untrained (Section 8).

3.2 The operative variable

For the two retrieval tasks the input position j^* that holds the answer is fixed by construction. At the query position we read the attention weight on j^* , taken as the maximum over heads and layers, and call it the attention on the correct source. We take the maximum because the readout can

use the correct value if any head or layer routes it, and we verify in Section 4.1 that the comparison with the variance candidate does not depend on this choice. This is the direct measurement of the property we hypothesize to be operative.

3.3 The variance candidate and its intervention

We also track the attention-output variance, the per-token variance of $z = W_O \text{Attn}(x)$, per layer against length. The intervention associated with this candidate is a layer normalization on the attention output (Li, Boduljak, and Zhou 2025). The baseline residual update is $h \leftarrow h + z$, and the intervention replaces it with

$$h \leftarrow h + \text{LN}(z), \quad (1)$$

which holds the downstream variance of the attention output constant across length. We record the variance both before this normalization, which is the raw collapse, and after it, which confirms the statistic is held constant.

3.4 Model and training

The model is a four-layer pre-norm decoder transformer with model width 256, eight attention heads, and feed-forward width 1024. We train with AdamW, batch size 512, and a warmup-plus-cosine schedule. Training sequences have length 1 to 5, and we evaluate on a geometric ladder up to $50\times$ the training length. The model is small by design, because the variance collapse is most severe at this scale, so the comparison is run where the scale candidate is strongest.

3.5 Pre-registration

We fixed three rules before viewing outcomes. A cell is informative only if its training-length exact-match is at least 0.8, which excludes tasks the model never learned. An effect at long length is read at the longest ladder length where the baseline first drops below 0.9 per-token accuracy, so that a task which never fails cannot be misread as evidence. We require the variance collapse to be reproduced before interpreting the variance comparison. Each order-invariant cell uses four seeds and the order-dependent task uses two. The pre-registration and its amendments are released with the paper.

4 Results

4.1 Attention on the correct source predicts accuracy

Figure 1 plots attention on the correct source against length for the baselines, with per-token accuracy overlaid. Attention on the correct source starts near 1.0 at short length and falls toward chance as length grows. Accuracy tracks it on every task and both positional encodings.

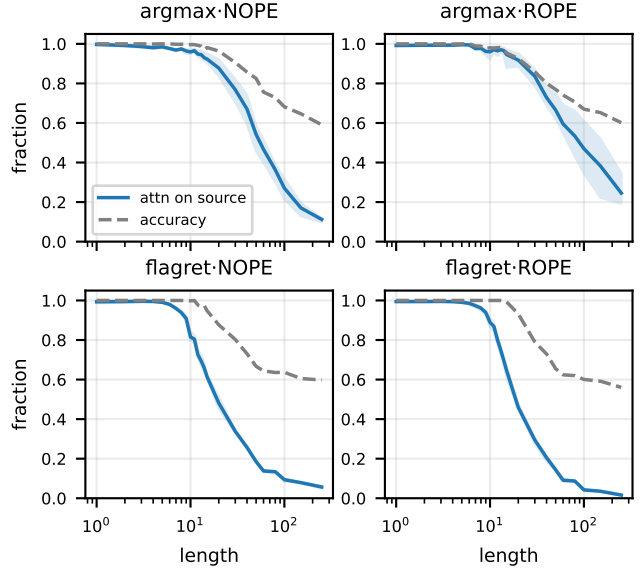


Figure 1: The operative variable. Attention on the correct source token (blue) falls with length, and per-token accuracy (dashed) tracks it, on both tasks and both positional encodings. Generalization holds while attention stays on the correct source and fails as it disperses.

Figure 2 makes the relation quantitative. Pooled over lengths and seeds, per-token accuracy is a tight function of attention on the correct source, with correlation 0.94 (left panel). Measured within each baseline cell and averaged, the correlation is 0.97. The pooled estimates use 384 length-by-seed points, with a bootstrap 95% interval of $[0.92, 0.95]$ for attention. The ranking does not depend on how layers are combined. Under mean-over-layers aggregation, which treats both statistics the same way, attention predicts accuracy with correlation 0.90 and the variance ratio with 0.33, and under last-layer aggregation the values are 0.91 and 0.22. The property we identified from the mechanics of retrieval is the one that predicts the behavior.

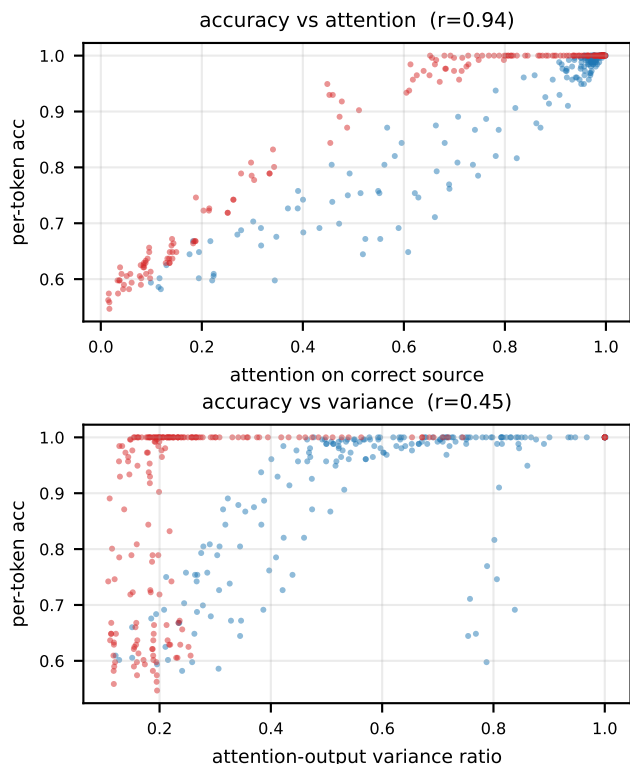


Figure 2: Baseline runs pooled over lengths and seeds. Top: accuracy against attention on the correct source, a tight relation with correlation 0.94. Bottom: accuracy against the attention-output variance ratio, diffuse with correlation 0.45, where flag retrieval (red) is a near-vertical band. Attention on the correct source is the better predictor.

We state the strength of this evidence plainly. That attention on the token a model must copy predicts whether it copies it is expected, so the relation is a localization of the failure and it is a starting point. The comparative content is what follows. We now measure a second quantity that also degrades with length and ask how well it predicts the same behavior.

4.2 A co-varying statistic: variance collapse

The attention-output variance collapses with length in every baseline. The variance of the deepest-collapsing layer, normalized to the training length, falls with length on every task, and the baseline ratio at $50\times$ lies between 0.27 and 0.48 (Table 1, and Figure S1 in the supplement). This statistic is a plausible candidate, since a distribution shift of this size affects every downstream component.

The variance is a weaker predictor of accuracy than attention on the correct source. In the pooled scatter (Figure 2, bottom) accuracy against the variance ratio has correlation 0.45, against 0.94 for attention. Within each cell the mean correlation is 0.59, against 0.97 for attention, and on flag retrieval the variance correlation is near 0.3. The variance moves with the failure and explains less of it.

4.3 Holding the variance constant does not restore generalization

The variance candidate names a concrete intervention, so we can go past correlation and repair the statistic directly. The normalization works as intended (Figure S2 in the supplement). For the intervention runs the variance before the normalization still collapses, while the variance after it is held near constant across length, with a post-intervention ratio close to 1.0 in every order-invariant cell (Table 1).

Holding the statistic constant does not change the behavior. Every cell masters the training length with exact-match 1.00 and degrades as length grows (Figure S3 in the supplement). On the two order-invariant tasks the intervention lies at or below the baseline at every length. The paired per-seed effect at the $50\times$ break point is at most zero in every cell (Table 1). Aggregated across the sixteen order-invariant cell-seed pairs the effect is negative in fifteen, with a mean of -0.047 and a bootstrap 95% interval of $[-0.062, -0.031]$, and a two-sided sign test rejects no effect ($p < 0.001$). The accuracy curves are in the supplement (Figure S3), and the intervention lies at or below the baseline in every retrieval panel while addition fails from a short length in both conditions.

The supplement shows the sign and the trend directly (Figure S4). The intervention’s per-token effect starts near zero at short length and moves downward as length grows, and the downward trend is steepest under RoPE. Repairing the co-varying statistic does not transfer to the behavior, and under RoPE it costs accuracy at the lengths where it was expected to help.

4.4 The intervention lowers attention on the correct source under RoPE

Our account predicts the sign of the intervention’s effect through the operative variable. An intervention that helps should raise attention on the correct source, and one that costs accuracy should lower it. The supplement compares attention on the correct source with the intervention off and on (Figure S5). The intervention does not restore the attention at long length in any cell. Under RoPE it lowers the attention further, from 0.25 to 0.07 on argmax and from 0.016 to 0.010 on flag retrieval at $50\times$. The behavior follows the operative variable, and the cost under RoPE is consistent with a component placed after attention disturbing the rotary frequency structure (Hua et al. 2024).

4.5 The order-dependent contrast

Reversed addition is mastered at the training length and then fails at every extrapolation length, with exact-match 0 throughout (Figure S3 in the supplement, bottom row). Its failure does not reduce to dispersion off a single source, because the answer draws on many positions at once, so the retrieval account does not extend to it and we do not expect the same treatment to apply. The variance-stabilizing intervention raises per-token accuracy by 0.047 under NoPE while exact-match stays at 0, which shows a graded metric reading as progress where no sequence is solved. Under RoPE the intervention is neutral. No configuration recovers

Table 1: Order-invariant tasks (argmax retrieval, flag retrieval), four seeds. Per-token accuracy at multiplicative lengths, where $1\times$ is the training length. The variance column is the deep-layer collapse ratio at $50\times$ before and after the variance-stabilizing intervention; a value near 1.0 after the intervention means the statistic is held constant. The final column is the intervention’s paired per-seed effect at the $50\times$ break point, with the range across seeds and the count of seeds in which it lowers accuracy. Every cell has training-length exact-match 1.00.

task	PE, intervention	$1\times$	$3\times$	$10\times$	$50\times$	var (before, after)	effect at $50\times$
argmax	NoPE, off	1.00	0.98	0.83	0.59	0.33, 0.33	
	NoPE, on	1.00	0.97	0.80	0.58	0.20, 0.99	$-0.013 [-.035, +.004]$, 2/4
	RoPE, off	1.00	0.95	0.77	0.60	0.27, 0.27	
	RoPE, on	1.00	0.96	0.67	0.57	0.45, 1.00	$-0.033 [-.066, -.008]$, 4/4
flagret	NoPE, off	1.00	0.94	0.67	0.60	0.47, 0.47	
	NoPE, on	1.00	0.88	0.62	0.52	0.49, 1.00	$-0.081 [-.109, -.043]$, 4/4
	RoPE, off	1.00	0.99	0.65	0.56	0.48, 0.48	
	RoPE, on	1.00	0.96	0.53	0.50	0.43, 1.00	$-0.061 [-.074, -.047]$, 4/4

usable length generalization on addition, which marks the boundary of the account.

4.6 The result holds at larger scale

We repeat the full comparison at eight layers and width 512, roughly eight times the parameters, and the three findings reproduce. Attention-output variance still collapses with length, with deep-layer ratios from 0.14 to 0.47, and the normalization still holds the downstream variance constant. Attention on the correct source still predicts accuracy with within-cell correlation 0.97, against 0.60 for the variance ratio, and the ranking again holds under mean-over-layers and last-layer aggregation. The variance-stabilizing intervention still gives no benefit, with a paired per-seed effect at $50\times$ of -0.039 and -0.021 under NoPE and -0.034 and -0.079 under RoPE on argmax and flag retrieval, negative in thirteen of the sixteen pairs (sign test $p = 0.02$). The account is present at this larger scale and is not an artifact of the smallest model.

5 Why attention concentration and variance separate

We make the relationship between the two quantities precise. At a query position let the attention weights on the n visible keys be $a = (a_1, \dots, a_n)$, with $a_j \geq 0$ and $\sum_j a_j = 1$, and let the value vectors be $v_1, \dots, v_n \in \mathbb{R}^d$. The attention output is the convex combination $z = \sum_j a_j v_j$, and we write j^* for the position that holds the correct source. The two statements below are properties of these expressions under simplifying assumptions on the values and logits. We use them to explain the measured relationships. We do not claim the trained network meets the assumptions exactly.

Proposition 1 (variance is a function of dispersion). *Model the value vectors as uncorrelated across positions with a common mean and a common per-coordinate variance σ^2 . Then the expected per-coordinate variance of the attention output is*

$$\text{Var}(z) = \sigma^2 \|a\|_2^2 = \frac{\sigma^2}{n_{\text{eff}}}, \quad n_{\text{eff}} := \frac{1}{\|a\|_2^2} \in [1, n]. \quad (2)$$

This identity is elementary, and the decay of representation variance as attention spreads has been noted in analyses of value aggregation and attention sinks (Li et al. 2026). We restate it here to make the dependence on dispersion explicit. The quantity n_{eff} is the participation ratio, the effective number of attended positions. It is 1 when attention sits on a single token, where $\text{Var}(z) = \sigma^2$, and it is n under uniform attention, where $\text{Var}(z) = \sigma^2/n$. Attention-output variance therefore collapses exactly as attention disperses, and $\|a\|_2^2$ does not depend on which position is correct, so the collapse happens for a right or a wrong selection alike. The one-line justification is that per coordinate $\text{Var}(\sum_j a_j v_j) = \sum_j a_j^2 \sigma^2$ by uncorrelatedness.

Proposition 2 (concentration decays with length). *Suppose the query-key logit of the correct source exceeds each of the other $n - 1$ logits by at least a gap Δ . Then the attention weight on the correct source satisfies*

$$a_{j^*} \geq \frac{1}{1 + (n - 1)e^{-\Delta}}. \quad (3)$$

For a fixed gap this bound tends to 0 as $n \rightarrow \infty$, and it stays bounded away from 0 only if the gap grows at least logarithmically, $\Delta \geq \log(n - 1) + c$ for a constant c .

Concentration on the correct source is the quantity the readout depends on, and Proposition 2 says it decays as competitors accumulate unless the model sharpens its logits at the rate $\log n$. That is the rate length-scaled logit sharpening supplies (Nakanishi 2025), which is why an intervention of that form is the way to hold concentration at longer lengths (Section 6). The justification is $a_{j^*} = e^{s_{j^*}} / \sum_j e^{s_j} \geq e^{s_{j^*}} / (e^{s_{j^*}} + (n - 1)e^{s_{j^*} - \Delta})$.

The two propositions share one driver and control different things. A layer normalization sends $z \mapsto \gamma(z - \bar{z}) / \sqrt{\text{Var}(z)}$, which by Proposition 1 rescales the output by a factor set by $\|a\|_2$ and leaves the weights a unchanged. It restores the size of z and does not raise a_{j^*} , so by Proposition 2 it does not change which value dominates the readout. Variance collapse and selection failure co-vary because both follow from the growth of n_{eff} , and they separate because only a_{j^*} enters the selection. This is the account behind the measurements:

attention on the correct source predicts accuracy while the variance ratio predicts it less well, and holding the variance constant leaves the behavior in place.

The same structure explains the sign under RoPE. The normalization is a component placed after attention, and such components interact with the rotary frequency structure in ways that reduce attention concentration (Hua et al. 2024). In our setting that interaction lowers attention on the correct source (Figure S5 in the supplement), which is why the intervention moves from neutral under NoPE to a cost under RoPE. This refines the observation that a post-attention normalization can sharpen attention (Zsámboki et al. 2025), since under RoPE in our setting the effect on concentration reverses.

6 Restoring concentration

The account makes an interventional prediction. If attention on the correct source is the operative variable, then an intervention that raises it at long length should raise accuracy, where holding the variance constant does not. We test this directly with a length-scaled logit sharpening that grows the logit gap at the rate Proposition 2 requires (Nakanishi 2025), applied to the same trained models, and pre-registered before running.

The prediction holds, and it separates the two candidates cleanly (Figure 3). Sharpening raises attention on the correct source in three of the four cells, from 0.06 to 0.13 on flag retrieval under NoPE and from 0.02 to 0.12 under RoPE, and accuracy rises with it. The variance-stabilizing intervention raises neither, and its accuracy effect is at most zero in all four cells (Section 4.3). Across the four cells the accuracy gain and the attention gain move together with correlation +1.0 (Figure 3, bottom). The one cell where sharpening does not raise attention, argmax under RoPE, is the one cell where accuracy does not improve, which is the pattern the account predicts. Raising the operative variable raises the behavior, and raising the co-varying statistic does not.

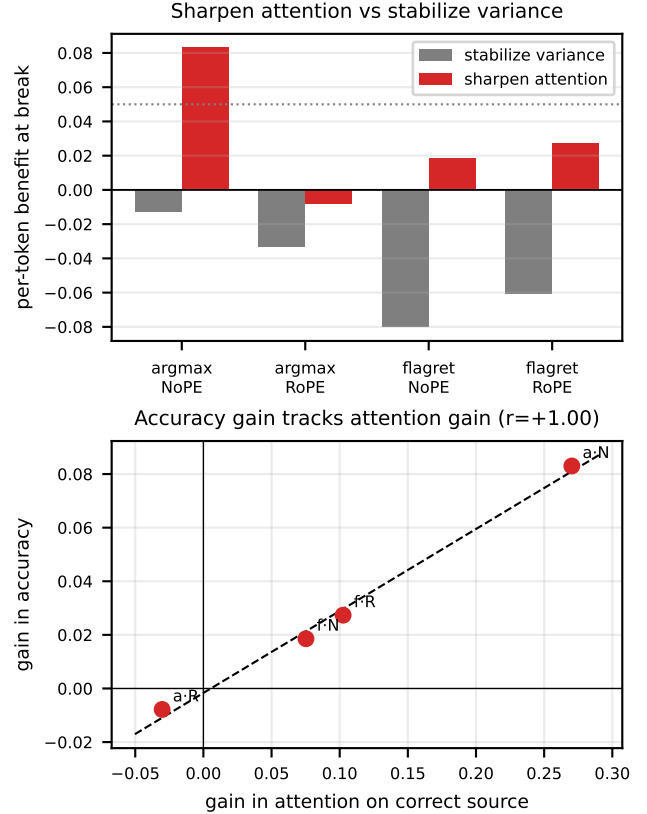


Figure 3: The intervention test. Top: per-token benefit at the break point for stabilizing variance (gray) and sharpening attention (red), by cell, with the dotted line at the pre-registered threshold of 0.05. Bottom: across the four cells the accuracy gain from sharpening tracks the gain in attention on the correct source, with correlation +1.0. Raising the operative variable raises accuracy, and the variance-stabilizing intervention raises neither.

We report the strength of the effect plainly. The single log-length lever is a partial rescue. Its per-token gain at the break point clears our pre-registered threshold of 0.05 on argmax under NoPE, where it reaches +0.083 across four seeds, and it is smaller on the other cells. A full recovery of short-length accuracy would need more than one inference-time knob (Vasylenko et al. 2025). The interventional result confirms the direction of the account, that attention selection is the operative lever, and it leaves the size of the achievable rescue to stronger concentration methods.

6.1 A direct manipulation

The sharpening intervention changes the attention logits, so it is an indirect test. We also manipulate the operative variable directly. At the answer-query position we overwrite the attention distribution to place a chosen mass on the correct source, holding the trained weights fixed, and read the accuracy. The construction and the pre-registration are released with the paper.

Forcing attention onto the correct source restores accuracy

(Figure 4, top). At $50\times$ the training length, where the baseline sits at 0.59 per token, patching the attention to attend fully to the source raises accuracy to 0.99, and the effect is monotone in the forced attention. Fifteen of the sixteen cells reach near 1.0, and the one exception is argmax under RoPE, the cell that also resisted sharpening. Accuracy rises with the forced attention on the source across the sweep at a held attention norm, with correlation 0.89 (Figure 4, bottom).

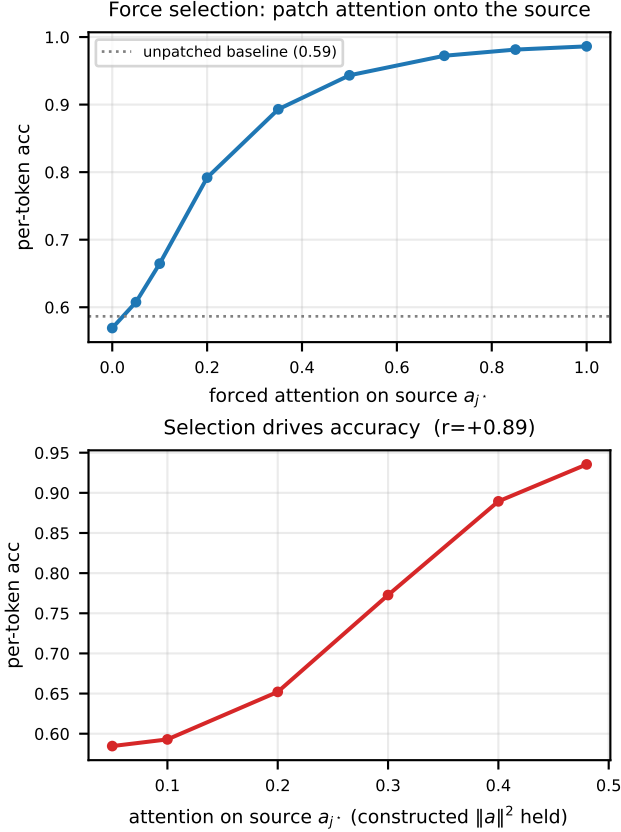


Figure 4: Direct manipulation of the operative variable, pooled over the sixteen models at $50\times$. Top: patching the answer-query attention to place mass on the correct source restores accuracy from the baseline (dotted) to near 1.0, monotone in the forced attention. Bottom: with the attention norm held by construction, accuracy rises with attention on the correct source (correlation 0.89).

We state the limits of this manipulation. It shows that attention on the correct source is sufficient to drive the readout, since forcing it restores accuracy and starving it returns accuracy to chance. It does not hold the attention-output variance exactly constant, because the source value vector is not exchangeable with the others, so the measured variance still drifts as the attention concentrates. The manipulation therefore demonstrates the sufficiency of selection, and we do not read it as a proof that the variance is causally inert.

7 Generalization to a pretrained language model

The results so far are on small transformers trained from scratch. We ask whether the operative variable appears in a real pretrained model. We probe Pythia-1.4B on in-context key-value recall: the prompt lists key-value pairs in the form `key: value`, a query repeats one key, and the model should produce the paired value. The value’s position is known, so we read attention on the correct source, and we vary the number of pairs as the context length. A large model routes retrieval through specific heads, so we identify the retrieval heads once as the eight heads with the highest attention from the query to the source at the shortest length, and measure those heads across the sweep.

The account appears in the real model (Figure 5). Accuracy and attention on the correct source both fall with context length, from 0.68 to 0.19 and from 0.48 to 0.22 over five to 160 pairs. Within a fixed length, attention on the correct source predicts which examples are correct, with a mean point-biserial correlation of 0.19, above the participation ratio $\|a\|_2^2$ at 0.12 and the attention entropy at 0.10. The per-example effect is modest, as expected in a model with many interacting heads, and the ranking matches the account, since attention on the correct source is the best of the candidates. The same quantity taken over all heads stays high across length, from 0.88 to 0.67, and tracks the failure poorly, which is why the retrieval heads are the place to read it.

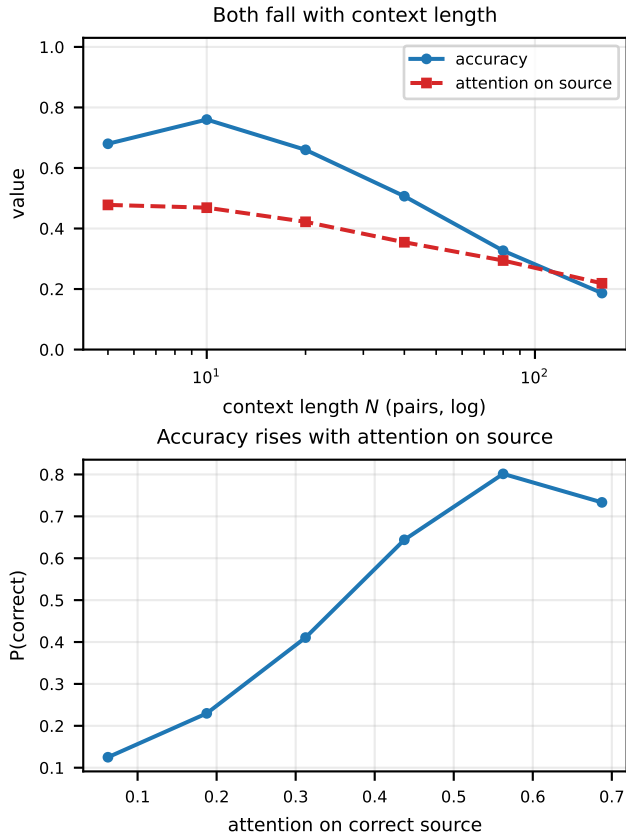


Figure 5: Pythia-1.4B on in-context key-value recall. Top: accuracy and retrieval-head attention on the correct source both fall as the context grows from five to 160 pairs. Bottom: pooled over examples, the probability of a correct answer rises with attention on the correct source.

7.1 Decomposing the failure: attention and readout

The co-decline localizes part of the failure to attention, and attention on the source is still non-trivial where accuracy is low, so we ask whether attention is the whole story. At the query token we read three things: whether attention reached the source (retrieval-head attention on the correct source), whether the value was retrieved into the residual stream (the correct value appears in the top ten of a logit lens at some layer, using the model’s own final norm and unembedding), and whether the value was output (the answer is correct). Among the failures at each length we separate two loci. A failure is attention-limited when the value is absent from the residual and attention on the source is low. A failure is readout-limited when the value is present in the residual yet the model does not output it, so a competing token wins after a successful retrieval.

The failure is not attention alone (Figure 6). At moderate length the failures are predominantly readout-limited, from 0.95 of failures at ten pairs to 0.58 at eighty, where the model retrieves the value and then loses it to a distractor. Attention-limited failures grow with length, from 0.05 to

0.60 over the same range, and become the majority only at the longest context. The two loci cross near the longest length we test. Attention on the correct source is therefore the dominant failure at the largest contexts, in agreement with the operative-variable account at the tail, while a downstream readout locus carries most of the failure at moderate length. The split is robust to the thresholds, with the readout share at the longest length between 0.34 and 0.50 across the ranges we tried, and the logit lens counts a value as present leniently, so the readout share is a lower bound.

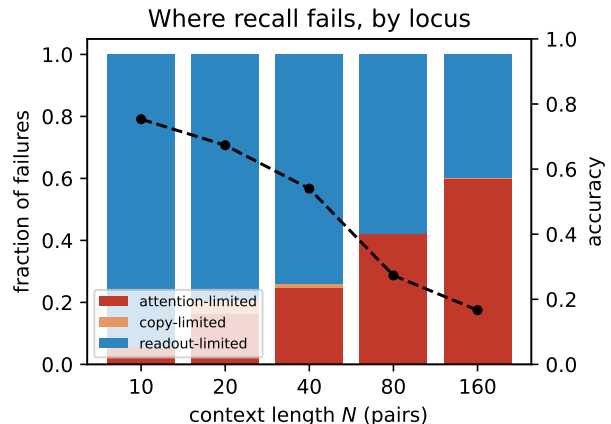


Figure 6: Where in-context recall fails in Pythia-1.4B, by locus, among the failures at each length. Readout-limited failures, where the correct value is in the residual yet not output, dominate at moderate length; attention-limited failures, where attention never reaches the source, grow with length and become the majority at the longest context. Dashed line: accuracy.

8 Limitations

The models are small. We report at four layers and width 256, and we replicate the full result at eight layers and width 512 (Section 4.6), which is roughly eight times the parameters, and we do not reach frontier scale. We do not observe a length-generalization improvement from holding the variance constant at either size, and we cannot exclude that a much larger model or a different task construction behaves differently.

Our identification of attention on the correct source as the operative variable rests on a direct correlation and a matched intervention. The intervention confirms the direction, since raising attention on the correct source raises accuracy in proportion, and it is a partial rescue that clears our pre-registered threshold in one of the four cells. The tracking correlation is measured over four cells, which makes it suggestive at the population level. A stronger concentration method would test how far the rescue can go. The account is stated for retrieval, and the order-dependent contrast shows it does not extend to tasks whose answer draws on many positions at once. The tasks are synthetic and chosen for control, so they isolate the mechanism and are not a measure of end-task benchmark performance. The pretrained-model probe is one model on one

retrieval task, and its within-example effect is modest, so it shows the account appears in a real model without measuring how large the effect is in general. The failure decomposition uses a logit lens, which reads intermediate residuals imperfectly and counts a value as present leniently, so the readout locus is a lower bound and the split is a characterization of this model and task. Our accuracy is measured with teacher forcing per token, an upper bound on free generation. Dictionary lookup did not reach our competence gate at this scale and is reported as untrained.

9 Conclusion

We asked which internal property supports transformer length generalization on retrieval, and we answered by direct measurement. Attention on the correct source token predicts accuracy with correlation 0.97, across tasks and positional encodings, and generalization holds while that attention stays concentrated and fails as it disperses. A natural alternative, the scale of the attention output, collapses with length as well, predicts accuracy less well, and does not transfer to the behavior when an intervention holds it constant, with a cost under rotary position embeddings. Raising attention on the correct source by inference-time sharpening raises accuracy in proportion, and the accuracy gain tracks the attention gain across cells, while stabilizing the scale raises neither. Patching the attention directly onto the correct source restores accuracy from 0.59 to near 1.0, which shows the selection is sufficient to drive the behavior. The same pattern appears in a pretrained language model, where retrieval accuracy and attention on the correct source fall together with context length, and a logit-lens decomposition there separates the failure into an attention locus that dominates at the longest contexts and a downstream readout locus that carries most of the failure at moderate length. A convex-combination argument explains the difference, since variance collapse follows from dispersion for any selection while concentration on the correct source controls the outcome. An internal statistic that co-varies with a failure is worth measuring against the property the task actually requires, because the two can move together and only one of them supports the behavior.

Reproducibility

The code, the full result files, the analysis scripts, and the pre-registration written before the outcomes are released with the paper. The transformers are trained in PyTorch on a single GPU, and the pretrained-model probe runs Pythia-1.4B through the Hugging Face Transformers library with eager attention on one GPU. Every run uses a fixed seed, the retrieval tasks use four seeds, and the tables report bootstrap intervals and a sign test.

References

Anil, C.; Wu, Y.; Andreassen, A.; Lewkowycz, A.; Misra, V.; Ramasesh, V.; Slone, A.; Gur-Ari, G.; Dyer, E.; and Neyshabur, B. 2022. Exploring Length Generalization in Large Language Models. ArXiv:2207.04901., arXiv:2207.04901.

Hua, E.; Jiang, C.; Lv, X.; Zhang, K.; Sun, Y.; Fan, Y.; Zhu, X.; Qi, B.; Ding, N.; and Zhou, B. 2024. Fourier Position Embedding: Enhancing Attention’s Periodic Extension for Length Generalization. ArXiv:2412.17739. Shows components after attention cause spectrum damage to RoPE, harming length generalization., arXiv:2412.17739.

Kazemnejad, A.; Padhi, I.; Ramamurthy, K. N.; Das, P.; and Reddy, S. 2023. The Impact of Positional Encoding on Length Generalization in Transformers. ArXiv:2305.19466. Finds the no-positional-encoding (NoPE) setting generalizes well., arXiv:2305.19466.

Li, R.; Boduljak, G.; and Zhou, J. 2025. On Vanishing Variance in Transformer Length Generalization. ArXiv:2504.02827, arXiv:2504.02827.

Li, S.; Jiang, K.; Sun, J.; and Hu, T. 2026. The Structural Origin of Attention Sink: Variance Discrepancy, Super Neurons, and Dimension Disparity. ArXiv:2605.06611. Value aggregation in self-attention produces a systematic variance discrepancy across positions., arXiv:2605.06611.

McLeish, S.; Bansal, A.; Stein, A.; Jain, N.; Kirchenbauer, J.; Bartoldson, B. R.; Kailkhura, B.; Bhatele, A.; Geiping, J.; Schwarzschild, A.; and Goldstein, T. 2024. Transformers Can Do Arithmetic with the Right Embeddings. ArXiv:2405.17399. Length generalization on addition via digit-position embeddings., arXiv:2405.17399.

Mударисов, T.; Burtsev, M.; Petrova, T.; and State, R. 2025. Limitations of Normalization in Attention Mechanism. ArXiv:2508.17821. As more tokens are selected, softmax attention drifts toward uniform selection., arXiv:2508.17821.

Nakanishi, K. M. 2025. Scalable-Softmax Is Superior for Attention. ArXiv:2501.19399. Scales attention logits with context length to counter softmax flattening., arXiv:2501.19399.

Su, J.; Lu, Y.; Pan, S.; Murtadha, A.; Wen, B.; and Liu, Y. 2021. RoFormer: Enhanced Transformer with Rotary Position Embedding. ArXiv:2104.09864. Introduces rotary position embedding (RoPE)., arXiv:2104.09864.

Vasylenko, P.; Pitorro, H.; Martins, A. F. T.; and Treviso, M. 2025. Long-Context Generalization with Sparse Attention. ArXiv:2506.16640. Uses alpha-entmax sparse attention to avoid dispersion and generalize to longer contexts., arXiv:2506.16640.

Wang, J.; Ji, T.; Wu, Y.; Yan, H.; Gui, T.; Zhang, Q.; Huang, X.; and Wang, X. 2024. Length Generalization of Causal Transformers without Position Encoding. ArXiv:2404.12224. Tunes attention-head temperature to improve NoPE length generalization., arXiv:2404.12224.

Zhai, S.; Likhomanenko, T.; Littwin, E.; Busbridge, D.; Ramapuram, J.; Zhang, Y.; Gu, J.; and Susskind, J. 2023. Stabilizing Transformer Training by Preventing Attention Entropy Collapse. ArXiv:2303.06296. Studies attention entropy as a quantity tied to training stability., arXiv:2303.06296.

Zhou, Y.; Alon, U.; Chen, X.; Wang, X.; Agarwal, R.; and Zhou, D. 2024. Transformers Can Achieve Length Generalization But Not Robustly. ArXiv:2402.09371. Length generalization is attainable but fragile to seed and data order., arXiv:2402.09371.

Zsámboki, P.; Levi, B.; Smith, D. A. J.; Kagalwala, M.; Kell, A.; Liechty, S.; and Wang, C. 2025. Post-Norm can Resharpen Attention. ArXiv:2510.08341. Studies how a post-attention normalization reshapes attention sharpness., arXiv:2510.08341.